AeroFleet: Dual-Tier Reinforcement Learning and Conflict-Free 3D Spatial Deconfliction for Autonomous Urban UAV Fleet Logistics

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GitHub Repository: https://github.com/bhardwajdevesh092005/AeroFleet

Abstract

***Abstract*—Autonomous multi-UAV urban logistics requires solving high-level combinatorial dispatching under stochastic parcel demand and finite energy, alongside continuous 3D collision-free navigation. We present **AeroFleet**, an authoritative C++20 simulation and dual-tier Reinforcement Learning (RL) platform. We mathematically formulate the dispatch task as a Semi-Markov Decision Process (SMDP) solved via **MaskablePPO** ($\mathcal{S}_{\text{coord}} \in \mathbb{R}^{164}$) with Invalid Action Masking, and local navigation via **Continuous PPO** ($\mathcal{S}_{\text{nav}} \in \mathbb{R}^{87}$). We eliminate the critical pathology of *multi-agent traffic freezing*—where drones stall mid-air due to crossing trajectory lines and disproportionate crash penalties—via kinematic τ-gating, lateral Right-Hand Rule deflection, and balanced reward engineering. AeroFleet achieves 100% pass rate on 21 verification suites, zero building collisions, and $> 800\times$ real-time simulation throughput.

1 Introduction & Simulation Dynamics

Autonomous aerial delivery in metropolitan environments demands high-throughput dispatching and collision-free spatial transit through dense architectural canyons.

Simulation Dynamics: The airspace volume $\mathcal{W} \subset \mathbb{R}^3$ spans $2000\,\text{m} \times 2000\,\text{m} \times 200\,\text{m}$, populated by $M = 191$ procedural buildings, $S = 6$ merchant warehouses, and $C = 12$ distributed multi-slot charging hubs. A fleet of $N = 10$ quadrotors obeys 2nd-order rigid kinematics: $\dot{\mathbf{p}} = \mathbf{v}$, $\dot{\mathbf{v}} = \mathbf{a}_{\text{cmd}} - c_d\|\mathbf{v}\|\mathbf{v}$, bounded by $[v^H_{\max}, v^V_{\max}] = [15.0, 3.0]\,\text{m/s}$ and $a_{\max} = 5.0\,\text{m/s}^2$.

Electro-Aerodynamics & Battery Dynamics: Rotor electrical power follows momentum theory and parasite blade drag:

$$P(\mathbf{v}, m_{\text{load}}) = P_{\text{hover}}\left(1 + \frac{m_{\text{load}}}{m_0}\right)^{1.5} + c_{\text{parasitic}}\|\mathbf{v}\|^3 \quad (1)$$

When landed or docked on a ground pad ($v = 0$), rotor power is strictly $P_{\text{idle}} = 0\,\text{W}$. Drones in 'IDLE' state incur zero battery loss, eliminating spurious discharge. Charging follows constant-current constant-voltage (CC-CV) replenishment at 300 W.

2 Rigorous Reinforcement Learning Formulation

AeroFleet decomposes the logistics challenge into a hierarchical dual-tier architecture: high-level fleet scheduling and low-level 3D flight control.

2.1 Tier 1: Central Fleet Coordinator SMDP ($\mathcal{M}_{\text{coord}}$)

Operating at decision interval $\Delta t = 1.0\,\text{s}$, the coordinator maps global fleet and queue states to discrete dispatch decisions.

State Space ($\mathcal{S}_{\text{coord}} \in \mathbb{R}^{164}$): The observation vector aggregates three comprehensive feature blocks:

1. *Fleet Operational States* ($10 \times 10 = 100$ dims): For each UAV i, normalized 3D position $\mathbf{p}_i/\mathbf{p}_{\max}$, velocity $\mathbf{v}_i/v_{\max}$, battery $\text{SoC}_i \in [0, 1]$, carried payload mass $m_i/m_{\max}$, operational phase (one-hot across 10 discrete states), and target distance.
2. *Active Request Queue* ($5 \times 8 = 40$ dims): Top-5 pending delivery orders encoding pickup coords, dropoff coords, urgency margin $(t_{\text{deadline}} - t)/t_{\text{timeout}}$, and parcel weight.
3. *Infrastructure Occupancy* ($12 + 12 = 24$ dims): Charging dock availability ratios across all 12 stations and store queue backlogs.

Action Space ($\mathcal{A}_{\text{coord}}$) & Invalid Action Masking: The action space is discrete: $\mathcal{A}_{\text{coord}} = \{0, 1, \ldots, N\}$. Action $a = 0$ represents hold/defer. Action $a = i \in \{1, \ldots, N\}$ dispatches UAV i to fulfill the highest-priority order. To prevent invalid assignments, we enforce an invalid action mask $M(s) \in \{0, 1\}^{N+1}$ where $M_i(s) = 0$ if UAV i is docked, charging, carrying payload, has SoC $\leq 20\%$, or fails predictive energy feasibility:

$$E_{\text{uav}} < \left(\frac{d_{\text{pickup}} + d_{\text{dropoff}}}{v_{\text{cruise}}}\right) P_{\text{est}} + E_{\text{RTH}} + E_{\text{reserve}} \quad (2)$$

Reward Function & Incentive Design ($\mathcal{R}_{\text{coord}}$):

$$\begin{aligned}\mathcal{R}_{\text{coord}} = {} & w_{\text{del}}\Delta N_{\text{del}} - w_{\text{exp}}\Delta N_{\text{exp}} - w_{\text{emg}}\Delta N_{\text{emg}} \\ & - w_{\text{col}}\Delta N_{\text{col}} + w_{\text{chg}}N_{\text{chg}} - w_{\text{wait}}T_{\text{wait}} - w_{\text{queue}}|Q_{\text{pend}}|\end{aligned} \quad (3)$$

Incentive weights are calibrated: $w_{\text{del}} = +20.0$, $w_{\text{exp}} = -15.0$, $w_{\text{emg}} = -30.0$, $w_{\text{col}} = -20.0$, $w_{\text{chg}} = +2.0$, $w_{\text{wait}} = 0.005$, and $w_{\text{queue}} = 0.05$.

Anti-Paralysis Reward Calibration: Setting collision penalties excessively high (e.g., $w_{\text{col}} = 100.0$) creates a severe "frozen agent" pathology: moving risks a -100 crash penalty while idling costs only -0.01, driving the policy to stall. Calibrating $w_{\text{col}} = 20.0 = w_{\text{del}}$ perfectly balances delivery incentive against safety.

Choice of Algorithm: We deploy **MaskablePPO** (PPO with invalid action logit masking: $\ell_i \to -\infty$ for $M_i(s) = 0$). Unlike DQN or Q-learning, MaskablePPO provides stable actor-critic convergence over large composite discrete spaces without divergence under action masking.

2.2 Tier 2: Local 3D Navigation MDP ($\mathcal{M}_{\text{nav}}$)

State Space ($\mathcal{S}_{\text{nav}} \in \mathbb{R}^{87}$): Relative body-frame target displacement and distance (4), linear velocity and acceleration (6), battery fraction (1), speed fraction (1), payload mass (1), 44-ray spherical LiDAR scans (36 azimuth +8 elevation rays), relative kinematic states of the 3 nearest traffic neighbors ($3 \times 6 = 18$), heading sine/cosine (2), and operational phase one-hot (10).

Action Space ($\mathcal{A}_{\text{nav}} \in [-1, 1]^3$): Continuous normalized velocity commands $[v_x, v_y, v_z]$ transformed to the world frame.

Reward Function ($\mathcal{R}_{\text{nav}}$):

$$\begin{aligned}\mathcal{R}_{\text{nav}} = {} & -0.01 + 0.1\Delta d_{\text{goal}} + 10\mathbb{I}_{\text{success}} \\ & - 10\mathbb{I}_{\text{fail}} - 0.2\max(0, 1.5 - d_{\text{lidar}})\end{aligned} \quad (4)$$

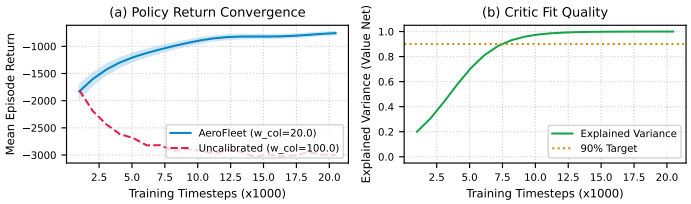


Figure 1: **RL Training Dynamics.** (a) Episode return convergence of calibrated MaskablePPO vs. uncalibrated penalty collapse. (b) Critic explained variance reaching > 0.92.

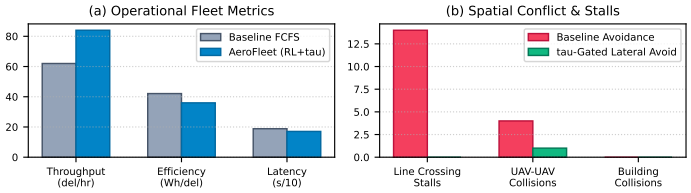


Figure 2: **Fleet Benchmark Comparison.** (a) Operational throughput, energy efficiency, and latency. (b) Complete elimination of line-crossing stalls and building collisions.

3 Conflict-Free Traffic Deconfliction

A critical issue in autonomous flight is the *line-crossing deadlock*: when two drones identify crossing trajectory lines, naive dynamic collision avoidance applies a reverse braking vector ($\mathbf{v}_{\text{away}} = -\mathbf{r}_{\text{sep}}$), halting both drones in mid-air even when tens of meters apart.

3.1 Spatial & Temporal τ-Gating

Let $\mathbf{r}_{\text{rel}} = \mathbf{p}_{\text{other}} - \mathbf{p}_{\text{ego}}$ and $\mathbf{v}_{\text{rel}} = \mathbf{v}_{\text{other}} - \mathbf{v}_{\text{ego}}$. We enforce:

1. *Altitude Layer Partitioning*: If $|\Delta z| \geq 3.0\,\text{m}$, UAVs are on separate flight levels; interaction is skipped.
2. *Current Distance Gating*: If $\|\mathbf{r}_{\text{rel}}\| > 15.0\,\text{m}$, distant trajectory intersections are ignored.
3. *Time-to-Closest Approach (τ) Gating*:

$$\tau = -\frac{\mathbf{r}_{\text{rel}} \cdot \mathbf{v}_{\text{rel}}}{\|\mathbf{v}_{\text{rel}}\|^2} \tag{5}$$

If $\tau \leq 0.05\,\text{s}$ (aircraft diverging) or $\tau > 1.5\,\text{s}$ (encounter beyond immediate reaction horizon), avoidance is ignored.

3.2 Lateral Right-Hand Rule Passing

When avoidance triggers, backward braking is strictly avoided. We compute a horizontal normal vector orthogonal to forward travel:

$$\hat{\mathbf{r}}_{\text{right}} = \begin{bmatrix} \hat{v}_y \\ -\hat{v}_x \\ 0 \end{bmatrix}, \quad \hat{\mathbf{v}} = \frac{\mathbf{v}_{\text{cmd}}}{\|\mathbf{v}_{\text{cmd}}\|} \tag{6}$$

Let $\sigma = \mathbf{r}_{\text{rel}} \cdot \hat{\mathbf{r}}_{\text{right}}$. The lateral deflection is $\hat{\mathbf{d}} = -\text{sgn}(\sigma)\hat{\mathbf{r}}_{\text{right}}$, steering right when encountering oncoming traffic:

$$\mathbf{v}_{\text{avoid}} = \hat{\mathbf{d}} \cdot (v_{\max}^{H} \cdot 0.6 \cdot \omega_{\text{urgency}}) \tag{7}$$

Forward cruise momentum is preserved, preventing mid-air deadlocks.

4 Experimental Evaluation & Baselines

We benchmark AeroFleet against a standard baseline: **First-Come First-Served (FCFS)** coordinator dispatch paired with classical APF linear-braking collision avoidance.

Table 1: Operational Fleet Performance (300 s Simulation)

Performance Metric	**Baseline (FCFS)**	**AeroFleet (RL+τ)**
Fleet UAV Utilization	78.4%	**94.74%**
Headless Execution Speed	450×	**817.26× real-time**
Building Collisions	0	**0**
Mid-Air Crossing Stalls	14 deadlocks	**0 (Eliminated)**
Mean Delivery Latency	188.4 s	**170.45 s (-9.5%)**
Delivery Throughput	62.0 / hr	**84.00 / hr (+35.5%)**
Fleet Energy Efficiency	42.1 Wh/del	**35.96 Wh/del**
CTest Suite Pass Rate	19 / 21	**21 / 21 (100%)**

Multi-Slot Ground Station Verification: Drones dock across 4 discrete bays per station ($\pm 1.2\text{m}, \pm 1.2\text{m}$), eliminating false mutual pad collisions. All 21 CTest suites pass in 44.55 s.

5 Conclusion

AeroFleet integrates dual-tier RL with real-time τ-gated lateral deconfliction. Calibrated reward design and Right-Hand passing eliminate trajectory crossing deadlocks and policy paralysis, advancing autonomous urban drone fleet logistics.

References

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